

24PY901
Quantum Principles, Structures and Computing
Module 1 – Unit 2

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Assignment 4: Postulates of Quantum Mechanics

Mapping with Linear Algebra formalism

References: Griffiths, Chapters 3. Formalism, Appendices 1. Vectors, 2. Inner products, 3. Matrices, and 5. Eigenvectors and Eigenvalues.

Wavefunction postulate

- (1) State the wavefunction postulate of quantum mechanics.
- (2) Concept of **normalized** kets –

- (a) Polarization states of light.

Consider preparing a collimated beam of light with a given state of polarization as shown in Figure. 1. If the light is propagating along $+y$ with amplitude E_0 , wavenumber k , and angular frequency ω , then $|H\rangle = E_0 \hat{i} \exp(ky - \omega t)$, $|V\rangle = E_0 \hat{k} \exp(ky - \omega t)$ are the independent horizontal and vertical states of polarization respectively.

Find the normalized ket $|S3\rangle$ representing the state of polarization after passing through

- collimator ($|S0\rangle \rightarrow |S1\rangle$),
- linear polarizer $|L_{\frac{\pi}{2}}\rangle$ ($|S1\rangle \rightarrow |S2\rangle$), and
- linear polarizer $|L_{\frac{\pi}{4}}\rangle$ ($|S2\rangle \rightarrow |S3\rangle$).

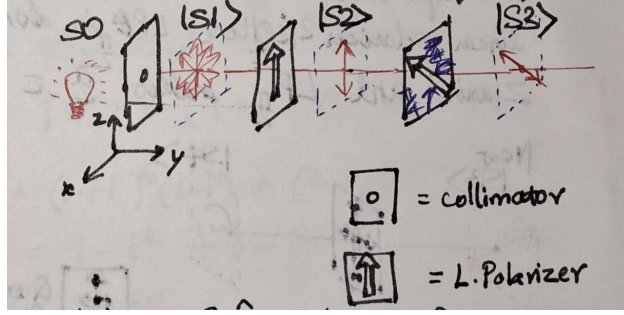


Figure 1: Polarized beam of light after passing through linear polarizers.

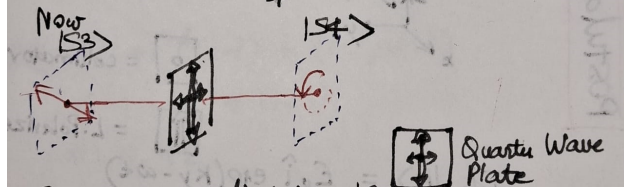


Figure 2: Polarized beam of light after passing through QWP.

Here, $|L_{\frac{\pi}{2}}\rangle$ is along z axis and $|L_{\frac{\pi}{4}}\rangle$ makes $\frac{\pi}{4}$ with respect to x -axis. Furthermore, the beam is now passed through a quarter wave plate (QWP) as shown in Figure. 2. Find the normalized ket $|S4\rangle$ representing the state of polarization after passing through QWP ($|S3\rangle \rightarrow |S4\rangle$).

(b) spin- $\frac{1}{2}$ system

(i) Normalize the ket

$$|\psi\rangle = (1 + i)|0\rangle + (1 - i)|1\rangle$$

(ii) Prove that the parametrized waveket

$$|\psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

has norm 1. Here θ, ϕ are real valued parameters.

(3) Concept of **orthogonalized** kets –

(a) Polarization states of light.

(i) If $|L_{\theta}\rangle = \cos \theta |0\rangle + \sin \theta |1\rangle$, then prove that $|L_{+\frac{\pi}{4}}\rangle$ is orthogonal to $|L_{-\frac{\pi}{4}}\rangle$.

(ii) If $|C_{+}\rangle = \frac{1}{\sqrt{2}} [|0\rangle + i|1\rangle]$, then find the circularly polarized ket $|C_{-}\rangle$ that is orthogonal to $|C_{+}\rangle$.

(b) spin- $\frac{1}{2}$ system

(i) A general ket for a spin- $\frac{1}{2}$ system is given by the two parameter form

$$|\psi(\theta, \phi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$$

Prove that $|\psi(\pi - \theta, \pi + \phi)\rangle$ is orthogonal to $|\psi(\theta, \phi)\rangle$.

Observable postulate

(1) State the observable postulate of quantum mechanics.

(2) From **action** of observable to operator –

(a) spin- $\frac{1}{2}$ system

(i) The observable $\tilde{S}_z$ has the following actions

$$\begin{aligned}\tilde{S}_z |0\rangle &= \frac{\hbar}{2} |0\rangle \\ \tilde{S}_z |1\rangle &= -\frac{\hbar}{2} |1\rangle\end{aligned}$$

Find the matrix representation of operator $\tilde{S}_z$.

(ii) The observable $\tilde{S}_x$ has the following actions

$$\begin{aligned}\tilde{S}_x |0\rangle &= \frac{\hbar}{2} |1\rangle \\ \tilde{S}_x |1\rangle &= \frac{\hbar}{2} |0\rangle\end{aligned}$$

Find the matrix representation of operator $\tilde{S}_x$.

(3) From operator to action of observable –

(a) spin- $\frac{1}{2}$ system

(i) The matrix representation of operator corresponding to the observable $\tilde{S}_y$ is

$$\tilde{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Find the action of $\tilde{S}_y$ on the computational basis.

(4) From operator to eigenvalues and eigenkets –

(a) spin- $\frac{1}{2}$ system

(i) Find the eigenvalues and eigenkets of the observables $\tilde{S}_z$ and $\tilde{S}_x$ whose matrix representations are given by

$$\tilde{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \tilde{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(5) From eigenvalues and eigenkets to operator –

(a) spin- $\frac{1}{2}$ system

- (i) The eigenvalues of observable $\tilde{S}_y$ are $\pm \frac{\hbar}{2}$, and eigenkets are

$$|\pm \frac{1}{2}_y\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

Find the matrix representation of observable $\tilde{S}_y$.

Time-evolution postulate

- (1) State the time-evolution postulate of quantum mechanics.
- (2) The Hamiltonian for a certain three-level system is represented by the matrix

$$\tilde{H} = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix}$$

where a, b , and c are real numbers.

- (a) If the system starts out in the state

$$|\mathcal{S}(0)\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

what is $|\mathcal{S}(t)\rangle$?

- (b) If the system starts out in the state

$$|\mathcal{S}(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

what is $|\mathcal{S}(t)\rangle$?

Measurement postulate

- (1) State the measurement postulate of quantum mechanics.
- (2) The Hamiltonian for a certain three-level system is represented by the matrix

$$\tilde{H} = \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Two other observables, A and B , represented by the matrices

$$\tilde{A} = \lambda \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \tilde{B} = \mu \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

where ω, λ , and μ are positive real numbers.

- (a) Find the eigenvalues and eigenkets of $\tilde{H}$, $\tilde{A}$, and $\tilde{B}$.
(b) Suppose the system starts out in the generic state.

$$|\mathcal{S}(0)\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix},$$

with $c_1^2 + c_2^2 + c_3^2 = 1$. Find the expectation values (at $t = 0$) of H , A , and B .

- (c) What is $|\mathcal{S}(t)\rangle$? If you measured the energy of this state (at time t), what values might you get, and what is the probability of each? Answer the same questions for observables A and B .